

SAI Happy Farms Farm ERP — Release Notes / What's New

Version: 1.0

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Status: Official Release Documentation

Overview

This release focuses heavily on infrastructure hardening, robust data synchronization, and strict Role-Based Access Control (RBAC) implementations. The core changes guarantee absolute data privacy between independent farms and ensure mathematical consistency across master inventory ledgers.

1. Security Improvements

The system's security posture has been heavily audited and fortified (Verified via 179 passing assertions in the Security Test Suite).

- BOLA Prevention (Broken Object Level Authorization):
- Previous Behaviour: Read/write permissions were loosely coupled to UI hiding.
- Current Behaviour: Firestore Security Rules and backend middleware mathematically verify that the requesting user's UID is explicitly mapped to the `farmId` being requested.
- Verification: Unauthorized cross-farm writes and exports are actively rejected with an HTTP 403 `AUTHORIZATION\_DENIED`.
- Error Sanitization:
- Previous Behaviour: Unhandled 500 errors could expose internal stack traces or raw database query strings to the client.
- Current Behaviour: All backend errors are intercepted. Stack traces are stripped, and raw database strings (e.g., `auth/user-not-found`) are replaced with generic security-safe messages.
- Verification: Validated by `firebaseErrors.security.test.ts`.
- Sensitive Log Redaction:
- Previous Behaviour: API keys and passwords could be logged in plain text in production server logs.
- Current Behaviour: Automated middleware dynamically scrubs strings resembling passwords, API tokens, and idTokens from logs.
- Verification: Validated via automated tests ensuring log immutability and redaction.
- Password Validation Enforcement:

- Previous Behaviour: Weak passwords were systematically possible.
- Current Behaviour: Passwords for all roles (Farmer, Supervisor, Admin) are strictly rejected by the backend if they are less than 8 characters long.
- Production Console Stripping:
- Previous Behaviour: Diagnostic ``console.log`` statements existed in the production bundle.
- Current Behaviour: The Vite build process successfully strips out diagnostic logging from the production output while retaining essential ``console.error`` logs.
- Verification: Manual inspection of build bundle verified clean stripping.

2. Data Integrity Improvements

- Atomic Inventory Deductions:
- Previous Behaviour: Potential for overlapping race conditions when updating bird/feed master ledgers.
- Current Behaviour: Submissions use strict Firestore atomic transactions. If mortality exceeds live birds, or feed usage exceeds master stock, the transaction explicitly aborts with ``INSUFFICIENT_BIRDS`` or ``INSUFFICIENT_FEED``.
- Report Corrections (Version 2 Delta Engine):
- Previous Behaviour: Correcting a report could lead to double-deductions in master inventory.
- Current Behaviour: Correcting a report automatically calculates the delta between V1 and V2. Only the mathematical difference is subtracted/added to the master ledger, and the V1 report is safely archived in a ``/revisions/`` subcollection.
- Verification: Validated by Scenario Test T06.

3. Import / Revert Improvements

- Revert Batch Safety Mechanism:
 - Previous Behaviour: Batches could potentially be reverted without strict secondary confirmation.
 - Current Behaviour: The Revert Batch workflow implements a mandatory two-step validation lock.
1. An agreement checkbox must be checked.
 2. The exact, case-insensitive ``batchId`` must be entered into a text field.
- Verification: The confirm button remains explicitly disabled if either condition is unmet.
 - Farm Mapping Rejection:
 - Previous Behaviour: Import scripts could attempt to map ambiguous farms.
 - Current Behaviour: Unknown or ambiguous farm mappings are explicitly rejected during the import validation phase. The system refuses to silently guess farm assignments.

4. Known Limitations & Pending Verification

The following items were identified in legacy documentation but remain Not verified from the available project evidence in this specific release cycle:

3. Test Execution Statistics: Specific historical claims regarding exact test counts (e.g., "XLSX import: 42/42 passed", "Frontend: 163 tests passed") are omitted pending official UAT re-execution.
 4. Push / Email Notifications: Alert mechanisms are currently strictly in-app visual cards. External push notifications via FCM or Email are not implemented.
 5. Data Export: Direct CSV/Excel export functionality from the dashboards is not present.
4. Hindi Localization: The translation directory does not currently contain a verified `hi.json` file. Language support is limited to English, Kannada, Telugu, and Tamil.

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